الأكاديمية العربية للعلوم والتكنولوجيا والنقل البحري

Arab Academy for Science, Technology & Maritime Transport

Lecture 4 Handout: Addressing Modes

Types of Addressing Modes

1- Register addressing mode

Examples:

MOV Destination, Source

- MOV al, bl
- Mov ax, bx
- Mov Bx, Dx
- Mov Si, Ax
- Mov eax, ebx ; 32 bit registers

In Register Addressing Mode, operands are located within the microprocessor's internal registers. This mode is efficient, as no additional memory access is required.

- **Examples**:
  - `MOV AL, BL` – Transfers the contents of `BL` to `AL`.
  - `MOV AX, BX` – Transfers the contents of `BX` to `AX`.
  - `MOV EAX, EBX` – Operates on 32-bit registers, moving data from `EBX` to `EAX`.

Note: Certain syntax errors occur if the register sizes do not match, such as `MOV AL, BX` and `MOV AX, BL`.

2- Immediate addressing mode

Examples:

- Mov al, 0f2H = MOV AL, 11110010b
- Mov ax, 234Ah
- Mov al, 200
- Mov AL , 0AH ← value = 10 in hexadecimal
- MOV DS, 234fh ; syntax error :
- Mov ax, 234fh;
- Mov ds, ax
- Mov cx, 1234h ← Immediate
- Mov Ax, 122
- Mov Cl, 1001B

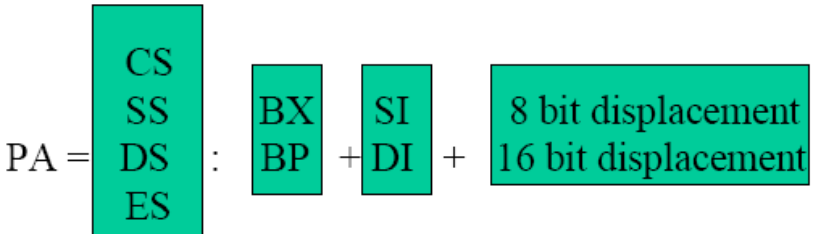
Immediate Addressing Mode includes a constant operand within the instruction itself, which is directly loaded into a register.

- **Examples**:
  - `MOV AL, 0F2H` – Loads hexadecimal `0F2` into `AL`.
  - `MOV AX, 234AH` – Loads hexadecimal `234A` into `AX`.
  - `MOV AL, 200` – Loads the decimal value `200` into `AL`.

Note: Attempting to load a segment register directly with an immediate value, such as `MOV DS, 234FH`, will result in an error. Instead, load the immediate into a general-purpose register first (e.g., `MOV AX, 234FH; MOV DS, AX`).

3- Memory Addressing Modes:

Memory Addressing Modes calculate a **Physical Address (PA)** based on a segment register and an offset.

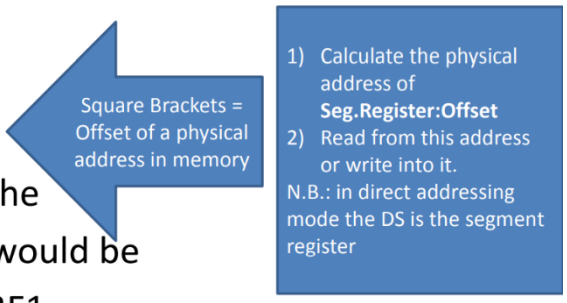


The most commonly used memory addressing modes are:

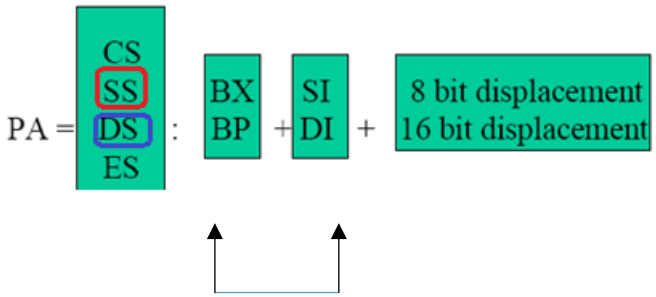
a. Direct addressing mode
b. Indirect addressing mode
c. Based addressing mode
d. Indexed addressing mode
e. Based-Indexed addressing mode

- **Direct Addressing Mode**
  - In Direct Addressing Mode, the offset address is specified within the instruction, and the default segment register is `DS`.
  - **Physical Address Calculation**: `PA = (DS * 10H) + offset`.
  - **Example**:
    - If `DS = 0200H`, `MOV CL, [1234H]` loads data from memory location `03234H` into `CL`.
    - For `MOV CX, [1234H]`, memory locations `03234H` and `03235H` are loaded into `CL` and `CH`, respectively.

Examples:
Mov Ax, [12F1]
If DS=2110 then the physical address would be 21100+12F1= 223F1



- **Indirect Addressing Mode**
  - This mode allows memory access through an offset stored in registers like `BP`, `BX`, `DI`, or `SI`.



  - When `BX`, `DI`, or `SI` are used, the `DS` segment is assumed. If `BP` is used, the `SS` (Stack Segment) register is assumed.

Example 1:

Instruction: `MOV AL, [BX]`

- **Assumptions**:
  - `BX = 1234H`
  - `DS = 0200H` (Data Segment is the default for `BX`)

Physical Address Calculation:

- **Formula**: `PA = (DS * 10H) + BX`
- **Calculation**: `PA = (0200H * 10H) + 1234H = 2000H + 1234H = 3234H`

The data at memory location `3234H` is loaded into `AL`.

Example 2:

Instruction: `MOV AX, [SI]`

- **Assumptions:**
  - `SI = 0456H`
  - `DS = 0300H` (Data Segment is used by default)

Physical Address Calculation:

- **Formula:** `PA = (DS * 10H) + SI`
- **Calculation:** `PA = (0300H * 10H) + 0456H = 3000H + 0456H = 3456H`

The contents at memory addresses 03456 and 03457 are transferred into the AX register.

The data is stored using little-endian convention, where the lower byte is placed in the lower address and the higher byte in the higher address, or vice versa depending on the endianness being used.

Example 3:

Instruction: `MOV CL, [BP]`

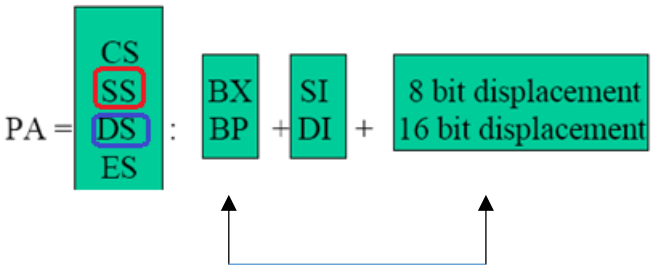
- **Assumptions:**
  - `BP = 0456H`
  - `SS = 0300H` (Stack Segment is used by default when addressing with `BP`)
- **Physical Address Calculation:**
  - **Formula:** `PA = (SS * 10H) + BP`
  - **Calculation:**

$$PA = (0300H * 10H) + 0456H = 3000H + 0456H = 3456H$$

- **Explanation:**
  The byte of data at memory address **3456H** (calculated physical address) is loaded into the `CL` register.

- **Based Addressing Mode** (`BX` or `BP`)

Based addressing mode uses a base register (`BX` or `BP`) combined with an optional displacement to calculate the effective address. `BX` uses the `DS` segment by default, while `BP` uses the `SS` segment.



Example 1:

Instruction: `MOV AX, [BX] + 4`

- **Assumptions:**
  - `BX = 0456H`
  - `DS = 0300H` (Data Segment is used by default with `BX`)
- **Physical Address Calculation:**
  - **Formula:** `PA = (DS * 10H) + BX + Displacement`
  - **Calculation:**

$$PA = (0300H * 10H) + 0456H + 4 = 3000H + 0456H + 4 = 345AH$$

- **Explanation:**

  The word of data at memory address **345AH** (calculated physical address) is loaded into the `AX` register.

  The contents at memory addresses 0345A and 0345B are transferred into the AX register.

little-endian convention: he data from memory address **0345A** is transferred to the lower byte of the AX register (**AL**), while the data from memory address **0345B** is transferred to the upper byte of the AX register (**AH**).

Example 2:

Instruction: `MOV DL, [BP] + 0010H`

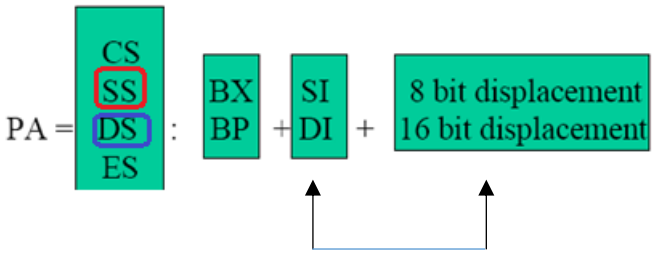
- Assumptions:
  - BP = 0456H
  - SS = 1234H (Stack Segment is used by default with `BP`)
- Physical Address Calculation:
  - Formula: $PA = (SS \times 10H) + BP + \text{Displacement}$
  - Calculation:

$$PA = (1234H \times 10H) + 0456H + 0010H = 12340H + 0456H + 0010H = 127A6H$$

- Explanation: The byte of data at memory address **127A6H** (calculated physical address) is loaded into the **DL** register.

• Indexed Addressing Mode (SI or DI)

Indexed addressing mode uses an index register (**SI** or **DI**) combined with a displacement to calculate the effective address. **SI** uses the **DS** segment by default, while **DI** can use either **DS** or **ES** depending on the instruction.



Example 1

- Instruction: `MOV AL, [SI] + 04H`
- Assumptions:
  - SI = 0123H
  - DS = 2000H (Data Segment is used by default with **SI**)
- Physical Address Calculation:
  - Formula: $PA = (DS \times 10H) + SI + \text{Displacement}$
  - Calculation:

$$PA = (2000H \times 10H) + 0123H + 04H = 20000H + 0123H + 04H = 20127H$$

- **Explanation:**
  The byte of data at memory address **20127H** (calculated physical address) is loaded into the **AL** register.

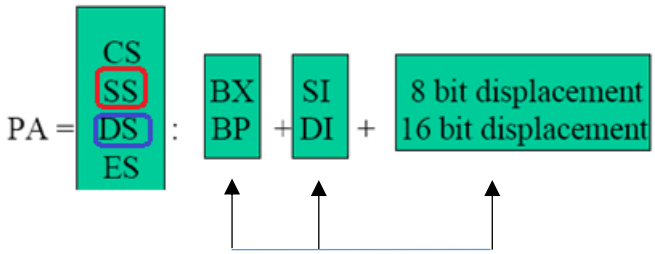
- **Example 2**
  - **Instruction:** `MOV BX, [DI] + 1234H`
  - **Assumptions:**
    - **DI = 0456H**
    - **DS = 3000H** (Data Segment is used by default with **DI**)
  - **Physical Address Calculation:**
    - **Formula:** $PA = (DS \times 10H) + DI + \text{Displacement}$
    - **Calculation:**

$$PA = (3000H \times 10H) + 0456H + 1234H = 30000H + 0456H + 1234H = 3168AH$$

  - **Explanation:**
    The word of data at memory address **3168AH** (calculated physical address) is loaded into the **BX** register.

- ## Based Indexed Relative Addressing Mode

  Based Indexed Relative Addressing Mode combines a base register (**BX** or **BP**) and an index register (**SI** or **DI**) along with a displacement to calculate the effective address. **BX** and **SI/DI** typically use the **DS** segment by default, while **BP** with **SI/DI** uses the **SS** segment.



Note: The instruction `MOV AX, [SI][DI] + displacement` is illegal. Similarly, the instruction `MOV AX, [BX][BP]` is also illegal.

Question:
For each of the following instructions, calculate the **physical address** and determine the data that would be loaded into the specified register. Show your steps for each calculation.

1. **Instruction:** `MOV AX, [BX + SI] + 08H`
   - **Registers:**
     - **BX = 0456H**
     - **SI = 0123H**
     - **DS = 2000H** (Data Segment is used by default with **BX** and **SI**)
2. **Instruction:** `MOV DL, [BP + DI] + 0020H`
   - **Registers:**
     - **BP = 0567H**
     - **DI = 0345H**
     - **SS = 1234H** (Stack Segment is used by default with **BP**)

Calculate the **physical address** for each instruction and identify what data would be loaded into **AX** for the first instruction and **DL** for the second.

- The answer to Part 1 of the question is as follows:
  - **Instruction:** `MOV AX, [BX + SI] + 08H`
  - **Assumptions:**
    - **BX = 0456H**
    - **SI = 0123H**
    - **DS = 2000H** (Data Segment is used by default with **BX** and **SI**)
  - **Physical Address Calculation:**
    - **Formula:** $PA = (DS \times 10H) + BX + SI + \text{Displacement}$
    - **Calculation:**

      $$PA = (2000H \times 10H) + 0456H + 0123H + 08H = 20000H + 0456H + 0123H + 08H = 20187H$$
  - **Explanation:**
    The word of data at memory address **20187H** (calculated physical address) is loaded into the **AX** register.

- The answer to Part 2 of the question:
  - **Instruction:** `MOV DL, [BP + DI] + 0020H`
  - **Assumptions:**
    - **BP = 0567H**
    - **DI = 0345H**
    - **SS = 1234H** (Stack Segment is used by default with **BP**)
  - **Physical Address Calculation:**
    - **Formula:** $PA = (SS \times 10H) + BP + DI + \text{Displacement}$
    - **Calculation:**

      $$PA = (1234H \times 10H) + 0567H + 0345H + 0020H = 12340H + 0567H + 0345H + 0020H = 12C0CH$$
  - **Explanation:**
    The byte of data at memory address **12C0CH** (calculated physical address) is loaded into the **DL** register.

More examples of addressing modes can be found in the reference book:

The 8086 Microprocessor by Mazidi.